SWEN 360 / CMPE 460 | Software Design and Engineering

EchoEats

Dynamic Pricing Food Waste Marketplace

[TEAM SUBMISSION]

Course SWEN 360 / CMPE 460	**Assignment** Assignment 1 — Agile Foundations
Sprint Sprint 1 - 4 Weeks	**Semester** Spring 2026

Team Members

Yousif Shakeeb — Scrum Master & Developer
Mohamed Elshamy — Developer
Rashid Bomtia — Developer
Hussain Yusuf — Developer

A. Team Deliverables

Section A.1 — Agile Process & Project Management

1. Agile Process & Project Management

Sprint 1 Progression Kanban Board

The Kanban board's progress on Sprint 1 in three sections; the start, the middle, and the end clearly illustrate the progress of all the tasks in all four columns.

Sprint Start — Week 1

BACKLOG 6 issues	TO DO 4 issues	IN PROGRESS 0 issues	DONE 0 issues
Feature Dynamic pricing engine **Rashid - Hussain · 8 SP**	**Feature** Dual registration system **Mohamed · 5 SP**		
Feature Food item listing + photos **Ali · 5 SP**	**Setup** GitHub repo setup **Yousif · 2 SP**		
Feature Location-based map search **Hussain · 5 SP**	**Req** Requirements document **Rashid - Team · 3 SP**		
Feature Reservation + hold system **Rashid . 3 SP**	**Setup** Tech stack selection **Mohamed**		

Design Wireframes — all screens **Yousif – Ali · 3 SP**			
Design Architecture diagram **Yousif – Ali - Mohamed · 3 SP**			

Sprint Middle — Week 2

BACKLOG 0 issues	**TO DO** 0 issues	**IN PROGRESS** 3 issues	**DONE** 3 issues
	Feature Dynamic pricing engine **Rashid - Hussain · 8 SP**	**Req** Requirements document **Rashid - Team · 3 SP**	**Setup** Tech stack selection **Mohamed**
	Feature Reservation + hold system **Rashid · 5 SP**	**Feature** Dual registration system **Mohamed · 5 SP**	**Design** Wireframes — all screens **Yousif - Ali · 3 SP**
		Feature Location-based map search **Hussain · 5 SP**	**Design** Architecture diagram **Yousif – Ali - Mohamed · 3 SP**

		Setup GitHub repo setup **Yousif · 2 SP**	**Feature** Food item listing + photos **Ali · 5 SP**

Sprint Middle — Week 3

BACKLOG 0 issues	**TO DO** 0 issues	**IN PROGRESS** 3 issues	**DONE** 3 issues
		Feature Reservation + hold system **Rashid · 5 SP**	**Design** Wireframes — all screens **Yousif - Ali · 3 SP**
		Feature Location-based map search **Hussain · 5 SP**	**Feature** Dual registration system **Mohamed · 5 SP**
		Feature Dynamic pricing engine **Rashid - Hussain · 8 SP**	**Setup** GitHub repo setup **Yousif · 2 SP**
		Design Architecture diagram **Yousif – Ali - Mohamed · 3 SP**	**Setup** Tech stack selection **Mohamed**

		Req Requirements document **Rashid - Team · 3 SP**	**Feature** Food item listing + photos **Ali · 5 SP**

Sprint End — Week 4

BACKLOG 0 issues	**TO DO** 0 issues	**IN PROGRESS** 0 issue	**DONE** 8 issues
			Feature Dual registration system **Mohamed · 5 SP**
			Feature Food item listing + photos **Ali · 5 SP**
			Feature Location-based map search **Hussain · 5 SP**
			Design Wireframes — all screens **Yousif - Ali · 3 SP**
			Feature Reservation + hold system

			Rashid · 5 SP
			Setup GitHub repo setup **Yousif · 2 SP**
			Req Requirements document **Rashid – Team · 3 SP**
			Feature Dynamic pricing engine **Rashid - Hussain · 8 SP**
			Design Architecture diagram **Yousif – Ali - Mohamed · 3 SP**
			Setup Tech stack selection **Mohamed**

Meeting Minutes — Sprint Planning

Date	Week 1 - 4
Duration	160 minutes
Attendees	All 5 Members of the Team
Sprint Goal	Provide the foundation for EchoEats, including dual registration, food listing, dynamic pricing engine prototype, location search, Payment Processes, and full design document.
Scrum Master	Yousif Shakeeb
Product Owner	Team members
Definition of Done	Code merges into main, unit tests pass, peer review by atleast 2 member or more.
Tech Stack	This project consists of developing a fully functioning website, along with web server hosting, a relational database with geolocation, a caching layer, and real-time communication features.

Selected User Stories & Task Breakdown

Story ID	User Story Summary	Assigned To	SP	Priority
ST-01	As a consumer, I want to see when the product offers will be finished, by displaying timers.	Rashid Bomtia	8	High
ST-02	As a user, I want detailed description of each product, when I click on it.	Mohamed Elshamy	5	High
ST-03	As a consumer, I want to be able to browse products with high quality images, and be able to view the products dimensions	Ali Al Buainain	5	High
ST-04	As a consumer, I want to be able to use the map location feature	Hussain Yusuf	5	Medium

	to see where the product is available			

Table 2 — Team Tasks

Task ID	Task Summary	Assigned To	SP	Priority
ST-05	As a Scrum Master, I will track Sprint progress over the project and develop kanban boards to track overall progress of our website development, also I will be taking tasks in designing our websites UI (User Interface) and human interaction, to ensure smooth operations.	Yousif Shakeeb	3	High
ST-06	Designing Wireframes for main website screens	Ali Al Buainain / Yousif Shakeeb	3	High
ST-07	Designing High-level system architecture diagram	Yousif Shakeeb / Mohamed El Shamy / Ali Al Buainain	3	High
ST-08	Seting up GitHub repository and branches to commit	Yousif Shakeeb	2	High
ST-09	Documenting Requirements document with user stories	Rashid Bomtia / All Team	3	High

Total Story Points Committed: 34 SP | Team Capacity: 5 members × 7 SP | Sprint Duration: 4 Weeks

Daily Stand-Up Logs (8 Meetings)

#1 Week 1	**Week 1 :** Sprint planning beginning, team tasks appointing.
	Feb 2: Mohamed: assigned to work on dual registration system, and tech stack selection. Rashid: assigned to work on reservation hold system and dynamic pricing engine. Ali: assigned to work on food item listing, and wireframes. Hussain: assigned to work on location-based map search, and dynamic pricing engine. Yousif: assigned to scrum master, and designing UI, and wireframes as well. **Feb 5:** Mohamed: researching on dual registration system, and tech stack selection. Rashid: researching on engine integration for dynamic pricing engine, and reservation hold system. Ali: researching on food item listing, and wireframes. Hussain: researching on location-based map search, dynamic pricing engine. Yousif: Researching on layout of website, and primary color to be used.
	Blockers: None.

#2 Week 1	**Week 2:** Beginning of progression in project, major start in operations.
	Feb 12: Mohamed: progress work in dual registration system, and first to start on architecture diagram. Ali: completed food items pricing. Hussain: proceeding in dual registration system. Yousif: creating GitHub repo and setting it up. Rashid: progress in documentation that is required and beginning in dynamic pricing engine. **Feb 16:** Mohamed: working and progressing in tech stacks selection. Contributed in architecture diagram. Ali: completed photos formatting, quality process, and wireframe contribution. Hussain: preparing to work on dynamic pricing engine, in its beginning stages. Yousif: completed architecture work and wireframe work. Rashid: pre work on reservation and hold system.
	Blockers: Mohamed: had problems in dual registrations – will contact owner. Ali: Photo quality error encountered – will contact owner.

#3 Week 2	**Week 3:** Advancement of core operations
	Feb 20: Rashid: documentation progress almost complete and polishing final touches on reservation hold system. Mohamed: prototype of architecture diagram released. Ali: contributed to architecture diagram in client, and database elements. Hussain: connecting, and testing location map-based search progress. Yousif: finalizing architecture design layout and providing fast responsive server to access with for the website to function. **Feb 24:** Rashid: required document almost completed, final revision only needed. Mohamed: dual registration system done, and tech stack completion final revision only needed. Ali: finished client, and database elements implementation. Yousif: completed wireframes for all screen pages.
	Blockers: None

#4 Week 2	**Yesterday:** All operations are complete, revisions checked.
	Feb 27: All: completed work, and all systems have been safely implemented. final Check done. **Mar 3:** All: final week meeting, discussing what to work on next, and what new features to possibly add.
	Blockers: None.

Sprint Retrospective

Date	Week 4, Day 2
Duration	60 minutes
Attendees	All 5 team members present

What Went Well ✓	What Didn't Go Well ✗	Actions for Sprint 2 →
GitHub workflow went will, and was done, no error encountered.	Uncertainity in demand multiplier formula delayed our sprint by two days.	Before Sprint starts we will clarify all ambiguous requirements with the Product Owner - 30 min Q&A session
All daily stand-ups were less than 15 minutes and showed us issues early on.	The image upload feature had been understated; due to configuration we moved this to Sprint 2.	Performance benchmarking is now included in our definition of done for data heavy features.

GitHub Repository

Repository URL	https://github.com/YMShakeeb/echoeats-web
Visibility	Public — instructor access granted
Branches	6
Commits	All 5 team members have active commits on the repository

Repository Folder Structure

Directory / File	Purpose
frontend/	Website frontend — all screens, navigation, and UI components
backend/	Web server — API routes, business logic, and service coordination
database/	Schema files, sample data, and geospatial index configuration

Directory / File	Purpose
pricing-engine/	Dynamic pricing algorithm — decay curves, demand multiplier, and price floor
docs/	Architecture diagram, wireframes, and requirements document
tests/	Unit and integration tests

A. Team Deliverables

Section A.2 — Initial Product Artifacts

Initial Product Artifacts

Requirements Document

User Stories

ID	As a…	I want to…	So that…	Priority	SP
US-01	Vendor	Enroll my task with title, description, type.	I can register surplus food on the platform.	Must Have	5
US-02	Consumer	Register your email, dietary preferences and location.	I get customized deals for food.	Must Have	3
US-03	Vendor	Mention a food item along with base price expiry time quantity photos.	My surplus food can be discovered and purchased by consumers.	Must Have	5
US-04	Consumer	Observe a real-time price that declines over time.	I have the ability to determine the perfect time for maximum savings.	Must Have	8

US-05	Consumer	Use a map with vendor pins to search for food nearby.	I am able to discover bargains in my desired budget.	Must Have	5
US-06	Consumer	Filter your listings by dietary tags such as vegan, halal, and gluten-free.	The only thing I see is food that decides to meet my dietary requirements.	Should Have	3
US-07	Consumer	Put a hold on an item by reserving it for 15 minutes.	My items are guaranteed while in transport to vendor.	Should Have	5
US-08	Vendor	Establish a base price that will never decrease below that level.	I uphold a minimum revenue level regardless of timing.	Should Have	3
US-09	Consumer	I get a push notification 1 hour before an expired item.	I don't regret last chance deals.	Could Have	3
US-10	Consumer	View my savings and achievements badges, and Scores.	I have the motivation to use EchoEats sustainably.	Could Have	3

Functional Requirements

ID	Requirement	Priority
FR-01	The system shall have two different registration flows, the Vendor registration flow and the Consumer registration flow, with fields in the respective role profiles.	High
FR-02	The system must automatically reduce each food listing’s price with time. This price reduction is based on how much time remains until expiry and to what extent the food item is being demanded. The re-evaluation shall take place every 15 minutes.	High
FR-03	The system should be able to support three vendor-selectable pricing curves such as simple decay, exponential decay and stepped decay.	High
FR-04	The consumer's screen shall automatically update the displayed price without requiring the consumer to reload the page.	High
FR-05	The System Shall Show Food Listings Within Proximity to Consumers Current Location. Only Vendors Within Chosen Distance Shall Be Shown.	High
FR-06	A food listing will accept up to 5 images, diet tags, allergen information, first price, expiry date and time and then quantity.	High
FR-07	The system shall have a 15-minute time-limited hold preventing double-booking when a consumer reserves an item.	Medium
FR-08	The system shall automatically remove expired listings from search and flag them as Expired in vendor dashboards.	Medium

Non- Functional Requirements

Category	Requirement
Performance	The site should load faster than 95 percent of other sites within 200 milliseconds as well as update prices, regardless of how many listings are open in less than 30 seconds. Search map results need to pop up in less than 1 second.
Security	A user's session will automatically end after one hour. Users must also log back into their account every seven days. Passwords are securely encrypted prior to being stored on the server. A few of the other ways that this system will protect user information are by encrypting all personally identifiable information such as user names, addresses, etc. That is stored in the system's database.
Scalability	The system must accommodate a total of 10,000 users at one time. The system also has to be able to send live price updates to a maximum of 5,000 users at any given moment. Additionally, all personal identifiable information like name, address, etc. That is stored in the database is encrypted. site has been designed with scalability in mind using caching to provide fast performance.
Usability	The website so it meets all standard accessibility guidelines. Also, the website must function properly on newer laptop computers made since 2020.

Initial design document

High level system architecture

Echo eats is a group of self-based posterior services that has a dedicated mobile application for the customer base it also includes a web board for retailers among other things it also includes specialized services to accommodate the different needs, as well as external providers for services with a three-layer storage system for the data.

Layer	Components	Technology
clients	All operating system apps/retailer web dashboard	Application (supporting all OS) and web browser UI
API	Rate limiting, SSL cancelation, load levelization, user authentication	A web server and a load balancer that handles all the requests that come in
Services	Reservations, pricing, food options, login, location service and reservation service	Separate services in the back that each do one thing
Data	Geospatial + Relational Database Cache + Pub/Sub Time-series history of prices	A main database that helps with finding locations, a database that caches information and another database that stores data over time
External API	Storage, updated notifications, payments, mapping	We use Google Maps for locations Stripe, for payments a service that sends push notifications and cloud storage for images
Real time running	Updated calculations for prices forwarded to clients	A live connection that works both ways and sends price updates to users away
Background tasks	The constant calculation timed every 15 mins	A scheduler that runs in the background and recalculates prices every 15 minutes automatically

Dynamic Pricing Algorithm

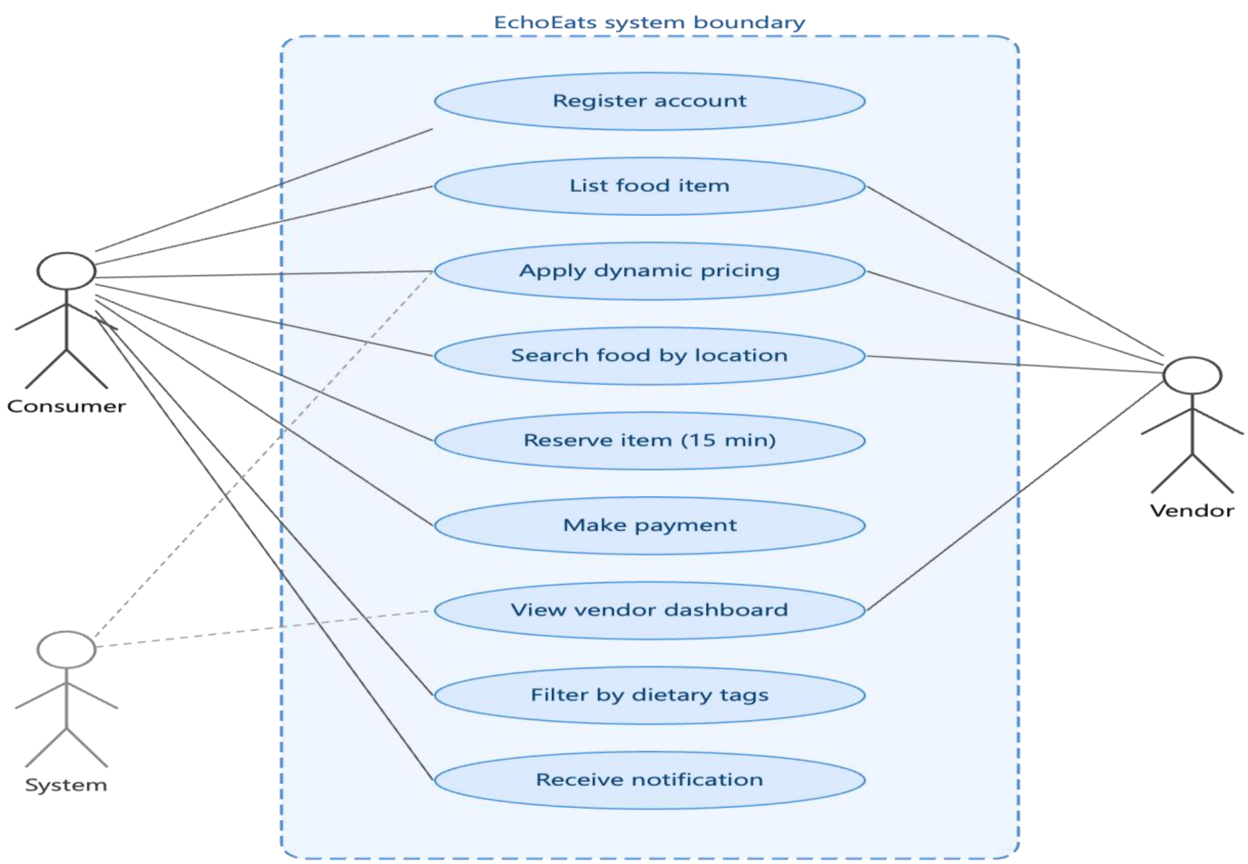
Core Formula	the price you pay is the price of the item multiplied by a few things like how much time is left and how many people want it.
Linear Decay	The price goes down at a rate from the start until it is time to stop selling the item so people can figure out what to expect
Exponential Decay	The price of the item stays close to the original price for most of the time it is for sale then it goes down really fast in the last hour, which makes people want to buy it before it is too late.
Stepped Decay	The price goes down at times for example it might go down to ninety percent of the original price then seventy percent, then fifty percent, then thirty percent, which is easy to understand.
Demand Multiplier	When a lot of people are looking at an item the price does not go down fast but when not many people are interested the price goes down faster to get people to buy it.
Price Floor	The person selling the item can set a price so the system will never sell the item for less, than that price.

UI wireframes – Main screens

Screen	Description	Key Elements
1. Home / Discovery	Default consumer view showing deals that are ending	Search bar, food cards with live price + countdown timer, filter chips
2. Item Detail	Full details for a chosen food listing	Photo, vendor name, distance, current price with a progress bar showing how much time is left Reserve or Save buttons
3. Map Search	Geospatial search view with vendor locations	Google Maps with grouped markers, list of vendors below the map filter, by distance
4. Vendor Dashboard	Vendor-facing analytics and listing management tools	Todays statistics including number of items sold and revenue earned active listings table Add New Listing button
5. Registration	Dual-path onboarding process for new users	Choose a role either Consumer or Vendor form fields to fill in and options to select dietary preferences

Screen	Description	Key Elements
6. Impact / Gamification	Consumer sustainability profile page	Number of meals saved amount of CO_2 saved, community rank and achievement badges

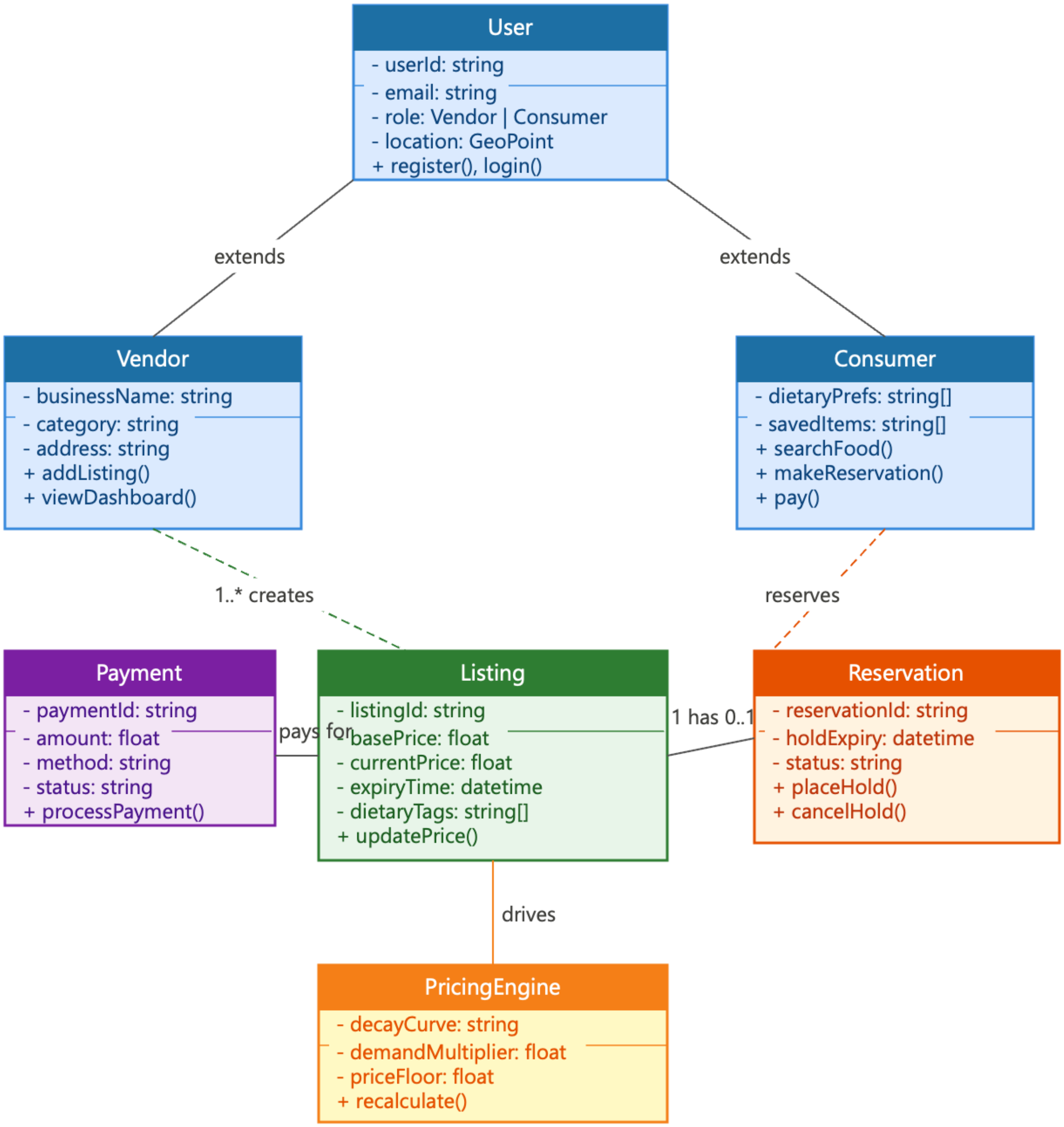
Use Case Diagram:



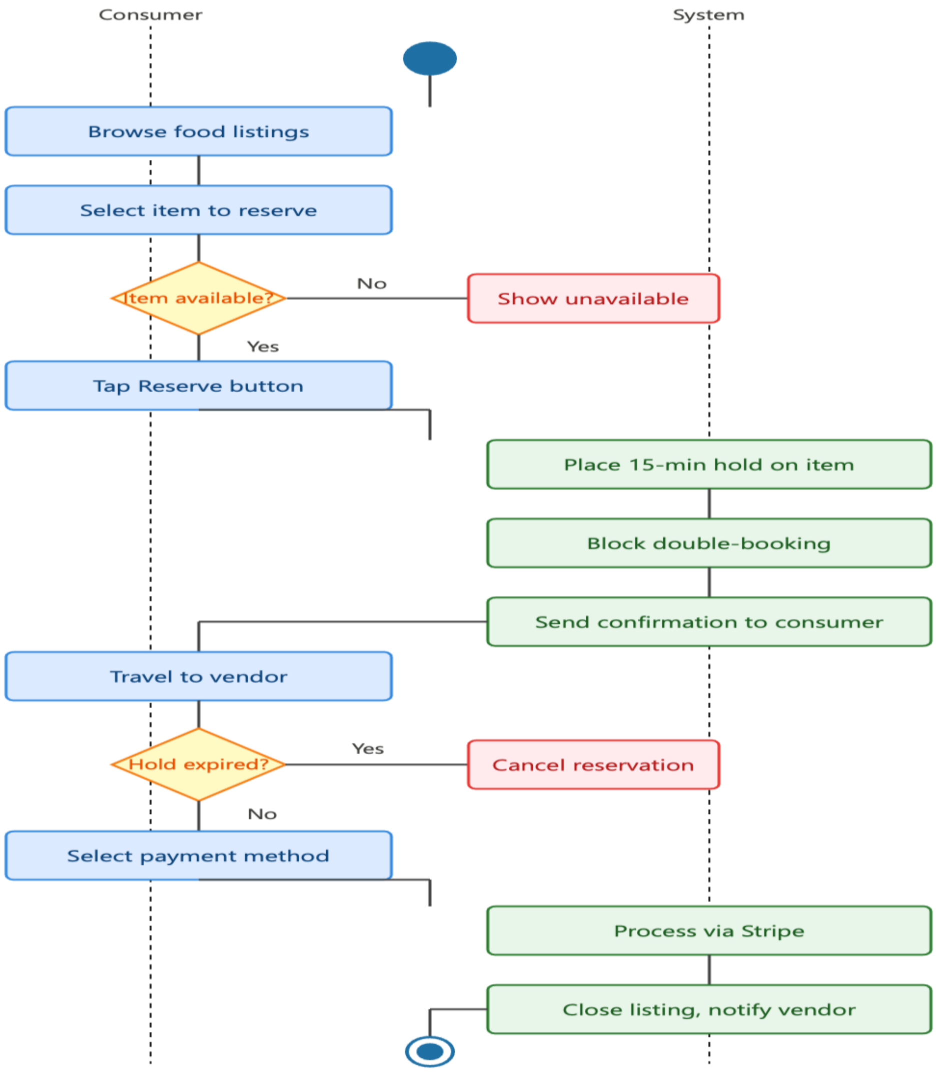
Use Case Descriptions:

Use case	Actor(s)	Precondition	Main flow	Postcondition
Registration	Consumer Vendor	User has no account	1. Select role (consumer/vendor) 2. Enter details & preferences 3. System validates & creates account	Account created, user logged in
List food item	Vendor	Vendor is logged in	1. Enter item name, base price, expiry 2. Upload up to 5 photos 3. Add dietary tags & quantity 4. Submit listing	Listing live; pricing engine starts
Dynamic pricing	System	Active listing exists	1. Every 15 min: recalculate price 2. Apply decay curve (linear/exp/stepped) 3. Apply demand multiplier 4. Enforce vendor price floor 5. Push update to consumers	All consumers see updated price
Reservation	Consumer	Item available, consumer logged in	1. Consumer taps Reserve 2. System places 15-min hold 3. Blocks double-booking 4. Timer shown to consumer	Item held; consumer travels to vendor
	Consumer Vendor	Item reserved or in cart	1. Consumer selects payment method 2. Stripe processes transaction 3. System confirms payment 4. Vendor notified; listing closed	Transaction recorded; listing removed
Search by location	Consumer	Consumer logged in with location	1. Consumer opens map view 2. System queries within radius (default 5 km) 3. Vendor pins shown on map 4. Consumer filters by dietary tags	Filtered listings displayed on map

Domain Class Diagram:



Activity Diagram:



INDIVIDUAL CONTRIBUTION REPORT

Yousif Shakeeb

EchoEats — Dynamic Pricing Food Waste Marketplace

Role	**Sprint**	**Marks Allocated**
Scrum Master	Sprint 1	4 Points

1. Facilitation of Agile Ceremonies

As Scrum Master for Echo Eats, I was responsible for organizing, facilitating, timeboxing all four of the Agile ceremonies throughout Sprint 1. My primary focus was that each ceremony should provide clear value, stay on point and aligned with the objectives and progress of the team.

Ceremony	Duration	Facilitation Notes
Sprint Planning	90 min	I enabled the Product Owner to clarify the priority of user stories in the backlog. Assisted the team in estimating effort in story points and ensured each story had a clear Definition of Done before acceptance to the sprint backlog.
Daily Stand-up	15 min/day	Facilitated daily stand-ups, where to keep up with the project using kanban board to track progress, and be able to manage
Sprint Review	45 min	Demonstrated live demonstration showcasing the functionalities accomplished through the sprint goal. I motivated the group to showcase the software that was functional while also obtaining the input from stakeholders in order to directly integrate the findings into the upcoming sprint backlog.
Sprint Retrospective	60 min	The Start-Stop-Continue model was used to organize the reflective process. Each of the action items had an owner assigned with a due date; this ensured that

Ceremony	Duration	Facilitation Notes
		any of the retrospective output created actionable results, rather than simply being observational.

2. Impediment Log

Below is a log of the impediments that were identified for Sprint 1, what actions were taken to address those impediments, and where each of these impediments stands now. As the Scrum Master, I followed each of these impediments from when they were first identified until when they were resolved.

#	Impediment	Steps Taken to Resolve	Raised By	Status
I-01	Unclear requirements for the Dynamic Pricing Algorithm Story led to uncertain estimates being made throughout the planning phase.	The Product Owner called an emergency Backlog Refinement Session where we broke down the Dynamic Pricing Algorithm Story into 3 smaller, well defined sub-tasks with specific Acceptance Criteria so that we could estimate each sub-task.	Scrum Master	**✓ Resolved**
I-02	A couple of team members had conflicting university exams halfway through the Sprint, therefore they were unable to work as much on their development tasks.	We rebalanced the Sprint Backlog by shifting some lower priority items to the remaining team members who were able to continue working. We also negotiated a Scope Reduction with the Product Owner to ensure that we met our Sprint Goal.	Team Members	**✓ Resolved**
I-03	Our Stand-Ups were consistently running	We added an open-parking-lot-board (visible timer) where any	Scrum Master	**✓ Resolved**

#	Impediment	Steps Taken to Resolve	Raised By	Status
	over the allotted time because the Status Updates from the team members would often drift into Problem-Solving Debates.	item from the standup was written down and then followed up at a dedicated 20 minute meeting right after the standup.		

3. Velocity and Progress Reflection

The team started the sprint well committed and completed the majority of all work about four of each five of their story-points were done as per definition of done. The only work that did not complete was the advanced pricing rule-configuration feature which was removed from scope of the sprint about half way through after we had a backlog-refinement-session and identified a blockage to this same feature.

34 Points Committed	**28** Points Delivered	**82%** Completion Rate

What Went Well

During this sprint, team members demonstrated effective collaboration and communication. The core marketplace feature development (i.e., User Registration, Restaurant Listing Management and the initial discount-trigger logic) were all completed within the allotted time frame. Additionally, all core features met the Definition of Done without requiring rework. A mid-sprint impediment led the team to implement a "parking-lot" practice which was quickly impactful as it decreased standup time and improved focus for their daily syncs.

Areas for Improvement

Story refinements need to take place sooner in the sprint cycle to keep away from the last minute changes in scope. Therefore, the team will begin holding story refinement sessions by day 3 of each sprint. This way, they can ensure that all stories are adequately sized and have clear acceptance criteria prior to starting the planning phase. As for capacity, each sprint should include consideration for the team members' academic commitments at the beginning of the sprint and not during it. This will help prevent the type of mid-sprint rebalancing they experienced during this sprint.

EchoEats — Scrum Master Individual Report — Sprint 1